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6 / 10 submitted

Q1. Selection Sort

Score: 10.00 **TC: 3/3 passed**

CODE

```
a = input("")

if " " in a:
    arr = list(map(int, a.split()))
else:
    arr = [int(x) for x in a]

for i in range(len(arr)):
    min_index = i

    for j in range(i + 1, len(arr)):
        if arr[j] < arr[min_index]:
            min_index = j

    arr[i], arr[min_index] = arr[min_index], arr[i]

print(arr)
```

INPUT
```
23 78 45 8 32 56
```

OUTPUT
```
[8, 23, 32, 45, 56, 78]
```

Q2. Insertion Sort

Score: 10.00 **TC: 3/3 passed**

CODE

```
a = input("")

if " " in a:
    arr = list(map(int, a.split()))
else:
    arr = [int(x) for x in a]

for i in range(1, len(arr)):
    key = arr[i]
    j = i - 1

    while j ≥ 0 and arr[j] > key:
        arr[j + 1] = arr[j]
        j -= 1

    arr[j + 1] = key

print(arr)
```

INPUT
```
8 5 7 1 9 3
```

OUTPUT
```
[1, 3, 5, 7, 8, 9]
```


Q3. Merge Sort

Score: 10.00 **TC: 3/3 passed**

CODE

```
a = input("")

if " " in a:
    arr = list(map(int, a.split()))
else:
    arr = [int(x) for x in a]

def merge_sort(arr):
    if len(arr) ≤ 1:
        return arr

    mid = len(arr) // 2
    left = merge_sort(arr[:mid])
    right = merge_sort(arr[mid:])

    result = []
    i = j = 0
```

```
    result = []
    i = j = 0

    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result += left[i:]
    result += right[j:]

    return result

print(merge_sort(arr))
```

INPUT

```
38 27 43 3 9 82 10
```

OUTPUT

```
[3, 9, 10, 27, 38, 43, 82]
```

Q4. Diagonal Matrix

Score: 10.00 TC: 3/3 passed

CODE

```
s = input().replace("[","").replace("]","").replace(" ","").split(",")
a = list(map(int,s))
n = int(len(a)**0.5)

f = True
for i in range(n):
    for j in range(n):
        if i != j and a[i * n + j] != 0:
            f = False
            break

print("Diagonal Matrix" if f else "Not Diagonal Matrix")
```

INPUT

```
[[1,2],[0,3]]
```

OUTPUT

```
Not Diagonal Matrix
```


Q5. Row and Column Sums

Score: 6.67 TC: 2/3 passed

CODE

```
s = input().replace(" ", "")
s = s[2:-2].split("],[")

a = []

for r in s:
    a.append(list(map(int, r.split(","))))

row = [sum(r) for r in a]
col = []

for j in range(len(a[0])):
    t = 0
    for i in range(len(a)):
        t += a[i][j]
    col.append(t)

print("Row= " + str(row))
print("Col= " + str(col))
```

INPUT

```
[[1,2],[3,4]]
```

OUTPUT

```
Row= [3, 7]
Col= [4, 6]
```

Q6. Maximum Subarray Sum

Score: 6.67 TC: 2/3 passed

CODE

```
a = list(map(int, input().split()))
m = s = a[0]

for x in a[1:]:
    s = max(x, s + x)
    m = max(m, s)

print(m)
```

Q5. Row and Column Sums

Score: 6.67 **TC: 2/3 passed**

CODE

```
s = input().replace(" ", "")
s = s[2:-2].split("],[")

a = []

for r in s:
    a.append(list(map(int, r.split(","))))

row = [sum(r) for r in a]
col = []

for j in range(len(a[0])):
    t = 0
    for i in range(len(a)):
        t += a[i][j]
    col.append(t)

print("Row= " + str(row))
print("Col= " + str(col))
```

INPUT

```
[[1,2],[3,4]]
```

OUTPUT

```
Row= [3, 7]
Col= [4, 6]
```

Q6. Maximum Subarray Sum

Score: 6.67 **TC: 2/3 passed**

CODE

```
a = list(map(int, input().split()))
m = s = a[0]

for x in a[1:]:
    s = max(x, s + x)
    m = max(m, s)

print(m)
```

INPUT

```
1 2 3 4
```

OUTPUT

```
10
```

Q7. Common Elements Among Three Lists

— Not Submitted —

Q8. Matrix Multiplication

— Not Submitted —

Q9. Sort Rotate Min Max

— Not Submitted —

Q10. Sort Matrix Search

— Not Submitted —